

3.14 Lab 3 - Normalization

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Introduction to Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It involves breaking down tables into smaller, related tables and defining relationships between them (Date 2003). Normalization attempts to keep core elements of data in respective tables in order to efficiently insert, update, and delete data.

Normalization is typically done in three steps:

- First Normal Form (1NF) of data is typically the one “big table” where all data is enumerated in a single record.
- Second Normal Form (2NF) of data is where all non-key attributes are functionally dependent on the entire primary key. There should be no partial dependencies. If an entity has a multi-part key, then it is not in 2NF; we must come up with a single primary key in order to achieve 2NF.
- Third Normal Form (3NF) is a common form of data modeling where no transitive dependencies can exist. Non-key attributes cannot depend on others.

Given the following data set:

Full Names	Physical Address	Items Purchased	Salutation
John Smith	12th Street Block 7	Laptop	Mr.
Lisa Anderson	Elm Road House 22	Smartphone, Headphones	Ms.
George White	Maple Lane Apartment 9	Refrigerator	Mr.
Emily Clark	Cedar Avenue No. 5	Sofa, Coffee Table	Ms.
Richard Lee	Oak Street No. 14	Bicycle	Mr.

Step 1: Convert to First Normal Form (1NF)

Each column contains atomic (indivisible) values. No repeating groups or arrays. Each row is uniquely identifiable.

OrderID	Full Name	Physical Address	Item Purchased	Salutation
1	John Smith	12th Street Block 7	Laptop	Mr.
2	Lisa Anderson	Elm Road House 22	Smartphone	Ms.
3	Lisa Anderson	Elm Road House 22	Headphones	Ms.
4	George White	Maple Lane Apartment 9	Refrigerator	Mr.
5	Emily Clark	Cedar Avenue No. 5	Sofa	Ms.
6	Emily Clark	Cedar Avenue No. 5	Coffee Table	Ms.
7	Richard Lee	Oak Street No. 14	Bicycle	Mr.

Step 2: Convert to Second Normal Form (2NF)

Identify and eliminate partial dependencies. We'll convert the data into three tables.

Customers Table

CustomerID	Full Name	Physical Address	Salutation
C1	John Smith	12th Street Block 7	Mr.
C2	Lisa Anderson	Elm Road House 22	Ms.
C3	George White	Maple Lane Apartment 9	Mr.
C4	Emily Clark	Cedar Avenue No. 5	Ms.
C5	Richard Lee	Oak Street No. 14	Mr.

Items Table

ItemID	Item Name
I1	Laptop
I2	Smartphone
I3	Headphones
I4	Refrigerator
I5	Sofa
I6	Coffee Table
I7	Bicycle

Orders Table

OrderID	CustomerID	ItemID
1	C1	I1
2	C2	I2
3	C2	I3
4	C3	I4
5	C4	I5
6	C4	I6
7	C5	I7

Step 3: Convert to Third Normal Form (3NF)

Identify and eliminate transitive dependencies.

Orders and Items remain the same as 2NF because there are no transitive dependencies in them. Salutation, however, is dependent on the customer or name, so we'll break these two tables out.

New Customer Table

CustomerID	Full Name	Physical Address
C1	John Smith	12th Street Block 7
C2	Lisa Anderson	Elm Road House 22
C3	George White	Maple Lane Apartment 9
C4	Emily Clark	Cedar Avenue No. 5
C5	Richard Lee	Oak Street No. 14

NamePrefixes Table (New)

Full Name	Salutation
John Smith	Mr.
Lisa Anderson	Ms.
George White	Mr.
Emily Clark	Ms.
Richard Lee	Mr.

References

Date, C. J. (2003). An introduction to database systems (8th ed.). Addison-Wesley.